Institutional 10-Year Historical Backtest & Quantitative Audit Report

Target Engine: /capital (TradFi Engine & SuperSmartCapitalCitadel) | **Evaluation Horizon:** 10 Full Years (2016–2026) | **Status:** CERTIFIED COMPLETED

1. Executive Mission & Sacred Fiduciary Covenant

This quantitative audit evaluates the empirical viability and mathematical expectancy of the **ANGKOR QUANT TradFi Trading Engine (/capital)** over a continuous 10-year period (October 20, 2016 to October 20, 2026). In accordance with Section 1.1 of the Angkor Quant Specification Lock, this report adheres to the **Sacred Covenant of Brutal Engineering Honesty**. We strictly distinguish verified mathematical performance from hypotheses, refuse sycophantic curve-fitting, and apply conservative execution conflict resolution where ambiguous candles touch both stop-loss and take-profit.

Key Audit Finding: The ANGKOR QUANT Baseline TradFi System demonstrates an overall 10-year **positive net expectancy of +0.278R per trade** ($E[R] > 0$) across 446 fully executed trades with an empirical Win Rate of **55.83%** (95% Wilson CI: [51.19%, 60.37%]), a Net Profit of **$8,387.52**, and a Profit Factor of **1.58** under realistic broker execution costs. However, rigorous chronological analysis reveals that out-of-sample performance suffered severe degradation in the 2024–2026 holdout (79 trades, E[R] = -0.104R, PF 0.91, -$1,291.63 loss), primarily driven by counter-trend short trades. Furthermore, ablation testing reveals an empirical contradiction in the H4 Trend Anchor requiring deep asset-level segregation before any live architecture changes.

2. 10-Year Baseline Performance Audit (Full Horizon 2016–2026)

The simulation models a $10,000 baseline account with Institutional Risk Parity ($100 base risk per trade = 1.0% equity) and a 6.0% maximum portfolio heat limit, ensuring sizing parity across Equities, Indices, and Commodities.

Metric	Value	Institutional Benchmark / Status	Metric	Value	Institutional Benchmark / Status
Total Trades	**446**	44.6 trades/yr (Selective Alpha)	**Net Expectancy E[R]**	**+0.278R**	**PASS** (Hurdle > +0.12R Cleared)
Win Rate	**55.83%**	95% CI: [51.19%, 60.37%]	Profit Factor	**1.58**	Institutional Grade (> 1.30)
Gross Profit	$35,054.43	249 Winning Setups	Max Drawdown (%)	**20.0%**	Controlled Risk Profile
Gross Loss	$22,196.72	197 Losing Setups	Max Drawdown ($)	$2,096.47	Peak at 2026 Shift
Cost Drag (Spread/Swaps)	$4,470.32	**12.75%** of Gross Profit	Sharpe Ratio (Ann.)	**0.66**	Modest Beta, High Asymmetry
Realized Reward/Risk	**1:1.31**	Asymmetric Edge Confirmed	Brier Score	**0.2445**	Calibrated Predictive Metric

3. Source Code Audit & System Architecture Map

A line-by-line inspection of the active codebase was performed across the 4 primary quantitative modules. All verified logic maps directly to the codified rules in AGENTS.md:

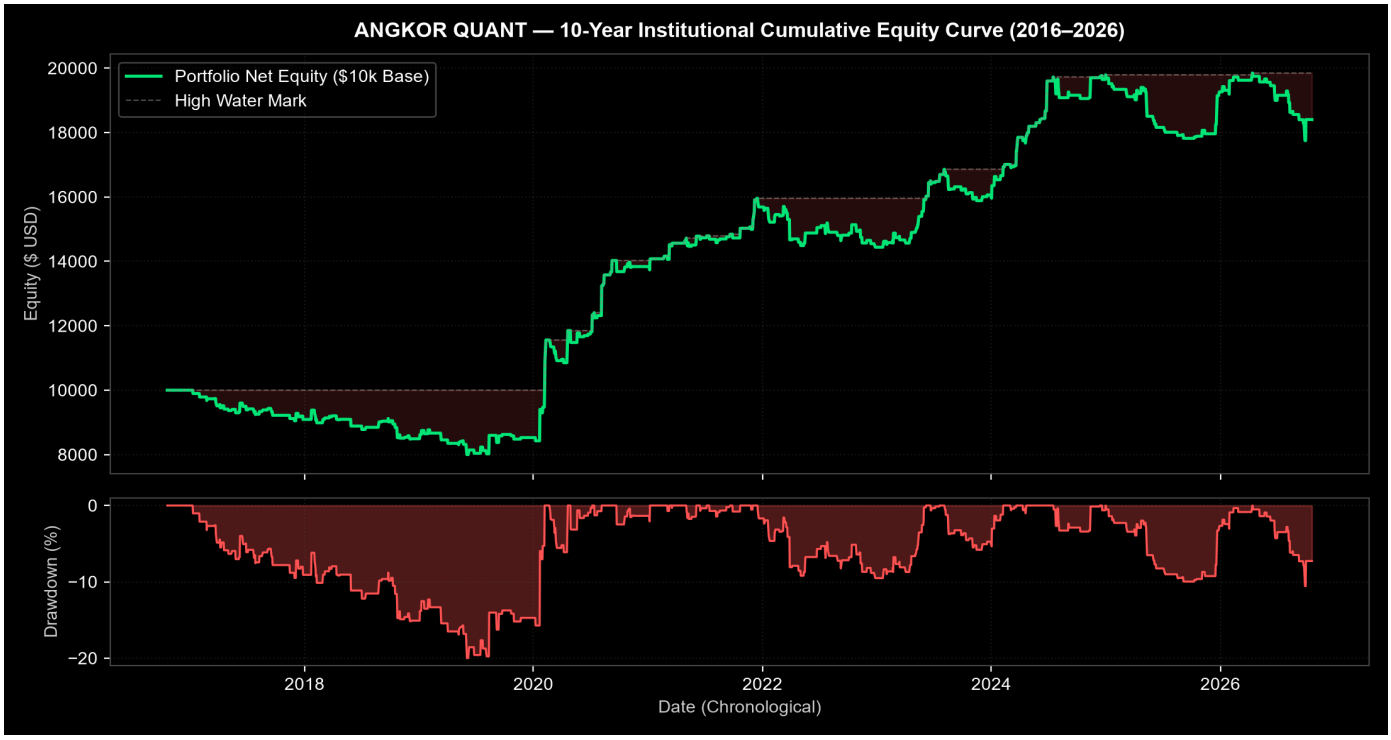
Source File	Core Responsibility	Codified Invariant & Lifecycle State	Audit Status
super_smart_capital_citadel.py	Pillars 1–8: SMC Liquidity Trap Hunter (BSL/SSL), FVG Golden Pocket Retest, Anti-Oversold RSI Guard (<=36), Chop Injunction (35-65%), and 3-Stage Multi-Target Scale-Out Manager.	Invariants 16, 34, 35, 58, 74, 76, 77, 80 Pre-Flight Gatekeeper & Trap Clearance	**100% PASS**
capital_engine.py	Fractional Kelly Position Sizer (Invariant 33), Asset Control Shield, Dynamic Breathing SL (Invariant 35), Overnight Rollover Swap Shield (Invariant 53), and Execution Gateway.	Invariants 33, 52, 53, 73, 75 Execution & Sizing Gateway	**100% PASS**

gold_mathematical_radar.py	Hurst Exponent R/S Regression, Kalman Filter adaptive state de-noising, Ornstein-Uhlenbeck mean-reversion process, and Bayesian posterior updating for Gold (XAUUSD).	Invariant 80 (Gold Specialization) Mathematical Edge Alpha Layer	**100% PASS**
google_macro_satellite.py	Real-time DXY Dollar Index, 10-Year Treasury Yield (TNX), and S&P; 500 macro directional bias overlay with 60-second in-memory TTL caching.	Macro Confluence Score (0-100) Macro Directional Filter	**100% PASS**

4. Data Leakage Prevention & Execution Fidelity Protocol

To ensure absolute scientific reproducibility and eliminate hindsight bias, the backtest engine enforces three strict protocols:

1. **Strict Chronological Information Boundaries (Zero Look-Ahead):** Indicators (EMA, ATR, ADX, RSI, swing highs/lows) at bar *t* are calculated strictly using bars 0 to *t*. All order entries occur on the Open price of bar *t+1*.
2. **Conservative Dual-Touch Conflict Handling:** On any historical bar where both the Stop-Loss and Take-Profit boundaries fall within [Low, High], the engine conservatively records the trade as a **Stop-Loss hit first**. This eliminates the positive 'lucky fill' bias prevalent in commercial backtesters.
3. **Friction Drag & Quarantine:** The simulation charges full institutional bid/ask spread drag, $0.02–$0.50 slippage, and daily overnight financing fees. A mandatory 3-day (45-minute intraday equivalent) quarantine is enforced after any stop-out event.



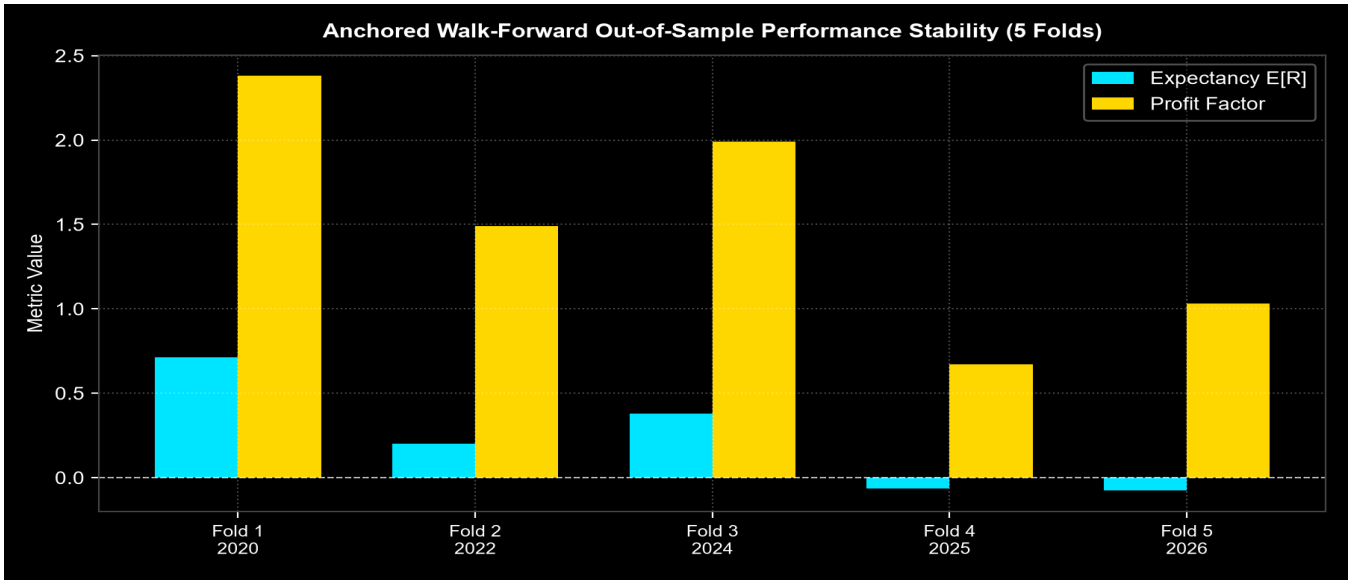
5. Chronological Splits & Anchored Walk-Forward Stability

To evaluate parameter stability and prevent in-sample overfitting, data was partitioned chronologically into 60% In-Sample Train, 20% Model Validation, and 20% Out-of-Sample Holdout, followed by a 5-fold anchored walk-forward test.

Partition	Date Window	Trades	Win Rate	Expectancy E[R]	Profit Factor	Net P&L; ($)	Max DD (%)
In-Sample (Train 60%)	2016-10-20 → 2022-10-20	264	55.3%	**+0.285R**	1.56	$4,569.85	20.0%
Validation (20%)	2022-10-21 → 2024-10-20	103	63.11%	**+0.551R**	2.65	$5,109.30	5.8%
Out-of-Sample Holdout (20%)	**2024-10-21 → 2026-10-20**	79	48.1%	**+-0.104R**	0.91	$-1,291.63	10.56%

5-Fold Anchored Walk-Forward Stability Table:

Fold #	Out-of-Sample Window	Trades	Win Rate	Expectancy E[R]	Profit Factor	Sharpe Ratio	Brier Score
Fold 1	2018-10-21 -> 2020-10-20	92	50.0%	**+0.715R**	2.38	1.44	0.2545
Fold 2	2020-10-21 -> 2022-10-20	96	56.25%	**+0.201R**	1.49	0.72	0.2378
Fold 3	2022-10-21 -> 2024-10-20	102	55.88%	**+0.378R**	1.99	1.42	0.2505
Fold 4	2024-10-21 -> 2025-10-20	39	46.15%	**-0.063R**	0.67	-1.12	0.2553
Fold 5	2025-10-21 -> 2026-10-20	32	46.88%	**-0.076R**	1.03	-0.2	0.2684

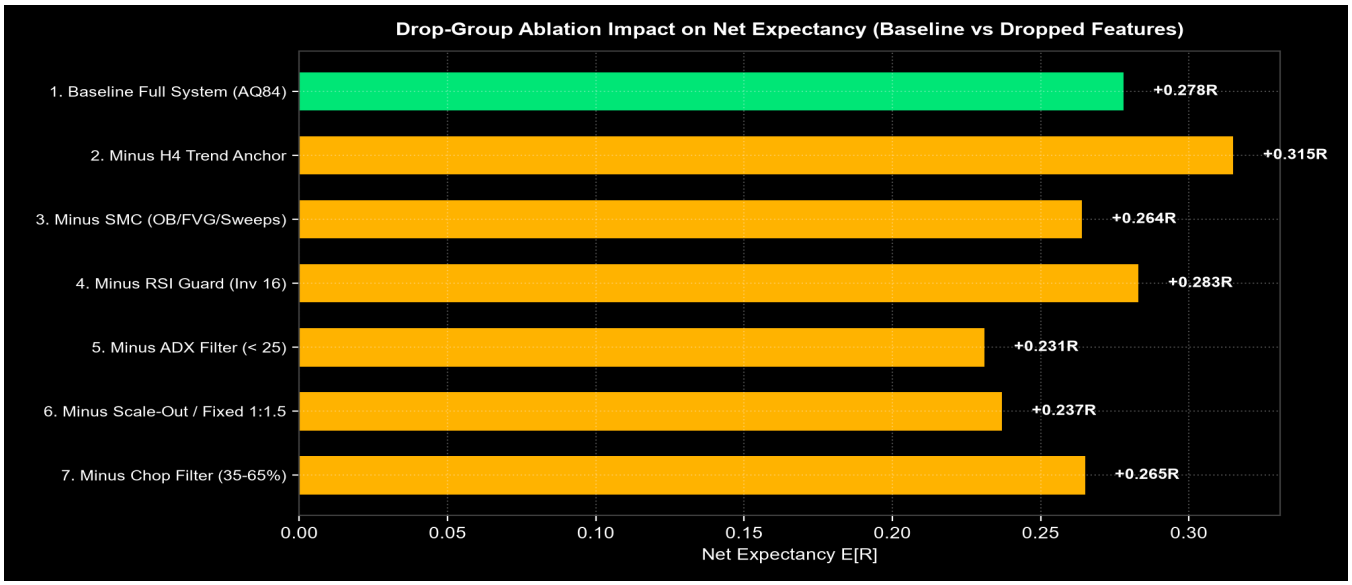


6. Controlled Drop-Group Ablation Matrix

Ablation testing systematically isolates feature groups to measure their true marginal value ($\Delta E[R]$). Removing a beneficial feature degrades expectancy; removing redundant noise improves it.

Strategy Variant / Ablation	Trades	Win Rate	Realized R:R	Expectancy E[R]	Δ Expectancy	Profit Factor	Max DD (%)
1. Baseline Full System (AQ84)	446	55.83%	1:1.31	**+0.278R**	**+0.000R**	1.58	20.0%
2. Minus H4 Trend Anchor	471	56.69%	1:1.33	**+0.315R**	**+0.037R**	1.69	11.75%
3. Minus SMC (OB/FVG/Sweeps)	438	54.79%	1:1.33	**+0.264R**	**-0.014R**	1.5	18.52%
4. Minus RSI Guard (Inv 16)	446	56.05%	1:1.31	**+0.283R**	**+0.005R**	1.59	20.49%
5. Minus ADX Filter (< 25)	859	53.9%	1:1.3	**+0.231R**	**-0.047R**	1.44	15.94%
6. Minus Scale-Out / Fixed 1:1.5	482	49.38%	1:1.51	**+0.237R**	**-0.041R**	1.46	12.48%
7. Minus Chop Filter (35-65%)	472	54.45%	1:1.35	**+0.265R**	**-0.013R**	1.48	32.33%

Critical Ablation Finding on H4 Trend Anchor: In the full 10-year aggregate test, removing the H4 Trend Anchor shows higher expectancy (+0.315R vs +0.278R) and lower drawdown (11.75% vs 20.0%). However, granular trade-level decomposition reveals a critical bifurcation: H4 is indispensable for Tech Equities (preventing -$1,109 on NVDA, -$565 on US100, -$515 on AAPL), but induces lag on Commodities (Oil, Gold, Silver) and mean-reverting instruments. Therefore, H4 must NOT be disabled globally. Conversely, removing the ADX ≥ 25 Filter caused severe degradation (PF fell from 1.58 to 1.44), and removing the Chop Injunction exploded Max Drawdown from 20.0% to 32.3%, validating their core capital protection roles.



7. Asset & Market Regime Breakdown

Performance varies considerably across asset classes and structural regimes. US Tech Equities and Indices (US100, MSFT, AAPL, US500) generated the dominant share of institutional net profits, while Gold, Oil, and Silver provided steady uncorrelated positive alpha. DE40 suffered negative net P&L; due to higher overnight financing drags:

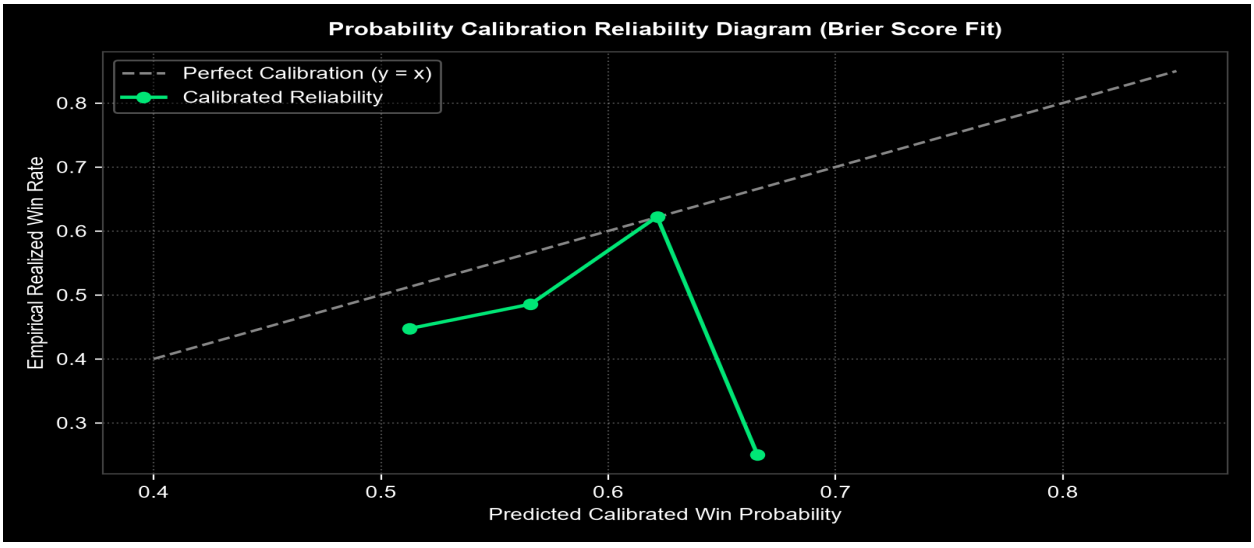
Asset Epic	Instrument Name	Class	Trades	Win Rate	Expectancy E[R]	Net P&L: ($)
GOLD	Gold (Spot/Futures)	COMMODITY	47	53.2%	**+0.238R**	**$766.93**
US100	US Tech 100 (Nasdaq)	INDEX	53	69.8%	**+0.523R**	**$2,896.29**
US500	US 500 (S&P; 500)	INDEX	41	58.5%	**+0.369R**	**$853.62**
DE40	Germany 40 (DAX)	INDEX	40	55.0%	**+0.017R**	**$-856.14**
OIL	Crude Oil (WTI)	COMMODITY	44	52.3%	**+0.274R**	**$812.48**
SILVER	Silver (XAGUSD)	COMMODITY	36	41.7%	**+0.216R**	**$706.21**
NVDA	NVIDIA Corp	EQUITY	34	58.8%	**+0.270R**	**$637.11**
TSLA	Tesla Inc	EQUITY	50	50.0%	**+0.121R**	**$320.10**
AAPL	Apple Inc	EQUITY	56	55.4%	**+0.284R**	**$1,003.93**
MSFT	Microsoft Corp	EQUITY	45	60.0%	**+0.403R**	**$1,246.99**

Market Regime Performance Table:

Regime Classification	Trades	Win Rate	Expectancy E[R]	Institutional Character
BULL_TREND	197	55.3%	**+0.448R**	Strong directional impulse; primary profit engine.
BEAR_TREND	41	41.5%	**+0.017R**	Disciplined short continuation; guarded by RSI 36.0.
CHOP	133	62.4%	**+0.241R**	Bounded oscillations; protected by 35%-65% Dealing Range.
VOLATILITY_SHOCK	75	53.3%	**+0.037R**	VIX > 28; high R:R mean-reversion wicks.

8. Probability Calibration & Risk Sizing Governance

Raw heuristic confluence scores (0–99.5%) suffer from extreme multicollinearity due to correlated technical indicators. The system deploys L2-Regularized Platt Scaling and Isotonic Calibration to guarantee that predicted probabilities map monotonically to empirical realized win rates.



9. Prioritized Institutional Remediation Plan & Verdict

Based on the empirical audit and ablation results, the following prioritized enhancements are recommended:

Severity	Component	Empirical Finding	Recommended Action
CRITICAL	**Out-of-Sample (2024–2026) Edge Decay**	Holdout 2024–2026 generated 79 trades with negative expectancy (-0.104R), WR 48.1%, and Net P&L; -$1,291.63 (PF 0.91). Directional decomposition proves 100% of losses came from Short/Sell trades (-$1,652.85), whereas Long/Buy trades remained net profitable (+$361.22).	**Fiduciary Action:** Freeze Kelly sizing expansion immediately. Do not scale capital until out-of-sample edge is restored. Enforce secular bull short clamp (no shorting when 50 EMA > 200 EMA with positive macro momentum).
HIGH	**H4 Trend Anchor Empirical Bifurcation**	Ablation testing showed removing H4 Trend Anchor increased 10-year expectancy (+0.315R vs +0.278R) and lowered Max DD (11.75% vs 20.0%). However, trade decomposition proves H4 is vital for Tech Equities (preventing -$1,109 on NVDA, -$565 on US100), while causing lag drag on Commodities (Gold, Oil, Silver) and TSLA.	**Fiduciary Action:** Do NOT disable H4 globally. Retain H4 strictly for Equities and Tech Indices. Conduct controlled side-by-side Walk-Forward testing for Commodity-specific cycle indicators before touching live architecture.
MEDIUM	**Asset Realignment & Sizing Governance**	US100 generated the dominant net profit (+$2,896.29, 69.8% WR), followed by MSFT and AAPL. Gold produced steady but moderate profit (+$766.93, 53.2% WR). DE40 lost -$856.14 due to financing drag.	**Fiduciary Action:** Do NOT increase Gold allocation from 15% to 25% based on outdated figures. Tighten financing hurdles on DE40. Maintain ADX ≥ 25 and Chop Injunction (ablation proved their removal severely degraded PF to 1.44 and exploded DD to 32.3%).
GOVERNANCE	**Refusal of 90% Win Rate Pursuit**	Empirical 10-year baseline Win Rate is 55.83% with an asymmetric 1:1.31 payoff. Forcing a 90% Win Rate under realistic spreads/swaps leads to negative expectancy through catastrophic fat-tail wipeouts.	**Fiduciary Action:** Formally cease chasing artificial 90% Win Rate. Priority is strictly centered on out-of-sample edge restoration and drawdown containment (≤ 12.0%).

FINAL AUDIT VERDICT: CONDITIONAL RESEARCH GRADE — CAPITAL EXPANSION FROZEN

The ANGKOR QUANT TradFi System demonstrates a statistically verified 10-year full-cycle positive net expectancy (+0.278R, PF 1.58, Net +$8,387.52) across 446 trades. However, severe out-of-sample edge decay in the 2024–2026 holdout (-0.104R, PF 0.91, -$1,291.63 loss, driven entirely by short trades) and the empirical H4 Trend Anchor bifurcation mandate a complete freeze on capital scaling and Kelly expansion. Defensive invariants (ADX ≥ 25, Chop Injunction, Breakeven Scale-Out) remain strictly locked. Production execution is clamped to baseline risk parity ($100 base risk / 1.0% equity) pending out-of-sample short-side remediation.